



Linux administration course

Practical Assignment#1: Linux Command Line Foundations

Instructor	Ins. Joshua K. IRADUKUNDA
Submission Deadline	September 19, 2026, 11:59 PM (Kigali time)
Submission Mode	Google Form (link to be shared separately by the instructor)
Total Marks	100
Individual Assignment	Individual. Every student submits their own account, recordings, and report.

1. Overview and Purpose

This assignment moves you from the theory of Linux into hands-on practice. By the end of it, you will have installed a working Linux environment, created and authenticated as your own dedicated system account, customized your shell prompt to display your identity, and worked through sixty of the commands a Linux administrator relies on most. You will finish by writing your own short shell script, because combining commands to solve a real problem is the clearest proof that you understand them.

2. Reference Material

This assignment is built entirely around “*The Linux Command Handbook: Learn Linux Commands for Beginners*” by Flavio Copes, published on freeCodeCamp:

1. **Handbook:** <https://www.freecodecamp.org/news/the-linux-commands-handbook/>
2. **Video tutorial:** <https://www.youtube.com/watch?v=ZtqBQ68cfJc>

Please, Read the handbook in full before you start recording.

3. Task 1: Environment Setup

Every task in this assignment depends on having a real Linux terminal to work in. Set one up as follows.

1. **Choose one option:** VirtualBox, VMware Workstation, Hyper-V (Windows Pro or Enterprise only), or WSL2 (Windows Subsystem for Linux). Any one of these is



acceptable, provided you end up with a working terminal running a genuine Linux distribution.

2. Download an ISO of Ubuntu Desktop or Server (recommended if this is your first time), or Kali Linux (recommended if you are already comfortable with Debian-based systems). Use only the official download pages, ubuntu.com or kali.org.
3. Create a new virtual machine and allocate at least 2 CPU cores, 4 GB of RAM, and 25 GB of disk space, then attach the downloaded ISO and complete the guided installation.
4. Once installed, open a terminal and confirm the installation:

```
lsb_release -a  
  
# or, if lsb_release is not available:  
  
cat /etc/os-release
```

5. Take a screenshot of this output for your report, together with your virtual machine's configuration screen and a screenshot of the completed installation.

Report deliverable for Task 1: two to three screenshots (VM configuration, installation completion, and OS version confirmation), each with a one- or two-sentence caption, plus a short note on your host machine's specifications and which distribution and version you chose.

4. Task 2: Create and Use a Dedicated Student Account

Administrators rarely work as root or as the default installation account. Every real system separates day-to-day work into named, accountable user accounts. You will create one for yourself.

1. Open a terminal on your new Linux system and confirm your current identity:

```
whoami
```

2. Create a new user account using your own Student ID as the username. For example, if your Student ID is 2026DEP123:

Common pitfall: Linux usernames are conventionally lowercase and, by default, cannot start with a digit. Since most Student IDs start with a number and include uppercase letters, the commands above may be rejected with an invalid-username warning.

Force your Linux to override the naming policy explicitly.

3. Set a password for the account when prompted, and, if you want to install software later, grant it sudo rights:

```
sudo usermod -sG sudo 2026DEP123
```



4. Switch into the new account, either with “su - 2026DEP123”, or by logging out of the installer account and logging back in as your Student ID account.
5. Confirm you are now operating as that account:

```
whoami
```

Report deliverable for Task 2: a screenshot of this whoami output showing your Student ID (or its lowercase form). This is the first proof-of-identity check in the assignment. Complete every remaining task while logged in as this account, not as root or the installer account.

5. Task 3: Customize Your Terminal Prompt

Your shell prompt is controlled by the PS1 environment variable, which understands a set of escape sequences: \u for your username, \h for the short hostname, \w for the current working directory (shown as ~ for your home folder), and \D{format} for the date combined with \t for the time.

Your prompt must match this exact layout:

```
(StudentID-⌘Hostname [YYYY-MM-DD HH:MM:SS])-[WorkingDirectory] >
```

For example, for Student ID 2026DEP123 on a machine called ThisPC:

```
(2026DEP123-⌘ThisPC [2026-09-13 16:59:38])-[~] >
```

```
(2026DEP123-⌘ThisPC [2026-09-13 16:59:38])-[~] >
```

“ThisPC” above is only an illustrative example hostname. Use whatever your own machine's hostname actually is (run “hostname” if you are not sure); you do not need to rename your machine to “ThisPC.” The circular symbol between your username and hostname is a stylized Unicode character popularized by Kali Linux's default prompt. It is not on a standard keyboard, so copy it directly rather than retyping it.

1. Make sure you are logged in as your Student ID account, then open your shell configuration file:

```
nano ~/.bashrc
```

2. Scroll to the end of the file and add this line exactly (copy the circular symbol rather than typing it):

```
export PS1='(\u-⌘\h [\D{%Y-%m-%d} \t])-[~] > '
```

4. Save the file (in nano: Ctrl+O, Enter, then Ctrl+X) and reload it so the change takes effect immediately:

```
source ~/.bashrc
```



5. Optional polish, worth trying but not required for full marks, adds colour:

```
export PS1='\[\e[1;32m\](\u-\h [D{%Y-%m-%d} \t])-\[w\]\[\e[0m\] > '
```

Report deliverable for Task 3: a screenshot of your terminal showing this prompt active. From this point on, every screenshot and every recording in this assignment must show this same customized prompt, so your instructor can confirm you were working from your own account throughout.

6. Task 4: Command Practice and Screen-Recorded Demonstrations

With your environment, account, and prompt ready, work systematically through all sixty commands from the handbook, in the order the handbook presents them, split across five videos of twelve commands each:

Video	Commands to cover, in this order
1	man, ls, cd, pwd, mkdir, rmdir, mv, cp, open, touch, find, ln
2	gzip, gunzip, tar, alias, cat, less, tail, wc, grep, sort, uniq, diff
3	echo, chown, chmod, umask, du, df, basename, dirname, ps, top, kill, killall
4	jobs, bg, fg, type, which, nohup, xargs, vim, emacs, nano, whoami, who
5	su, sudo, passwd, ping, traceroute, clear, history, export, crontab, uname, env, printenv

For every command in every video:

- Reproduce at least one example directly from the handbook, and explain out loud, and in captions, what it does.
- On each command, please go beyond the handbook: invent and demonstrate one original example of your own, using different files, directories, or arguments than the handbook uses, and explain why you set it up that way and what it proves you understand.

Recording requirements

- Screen-record your terminal. A webcam is optional and not required.
- Narrate or caption every command as you run it. A silent demonstration with no explanation earns no marks for that command.
- Your customized prompt from Task 3 must be visible throughout.



- Free tools you can use: OBS Studio, your operating system's built-in screen recorder, or Loom.
- Ten to fifteen minutes per video is a realistic target for twelve commands with explanations; there is no fixed maximum, but padding does not earn extra marks.
- Upload all five videos to your own Google Drive and set sharing to “Anyone with the link can view.” Test each link from a private or incognito browser window before you submit it.

7. Task 5: Build Your Own Shell Script

Choose at least five commands from the handbook, they do not need to come from a single video, and combine them into one original Bash script that solves a real problem you would actually want solved on your own machine. A script that only prints a greeting, or one copied from somewhere online, will not satisfy this task.

Ideas to spark your thinking (you are not limited to these)

- A daily disk usage and cleanup report for your home directory.
- A quick system health snapshot: uptime, memory, disk space, and logged-in users.
- An automated backup of a chosen folder into a timestamped archive.
- A log file monitor that reports the most recent errors seen.
- A simple account report for a system administrator.

Requirements

- The script must run correctly and produce a sensible, useful result.
- Comment the script clearly: state its purpose at the top, and next to each interesting line, explain what it does and why you wrote it that way.
- In your PDF report, include the full script text, a screenshot of it running successfully, and a short paragraph, five to eight sentences, explaining the problem it solves and why it matters to you.

9. Submission Guidelines

1. Finalize your five videos from Task 4.
2. Upload all five videos to Google Drive and set sharing to “**Anyone with the link can view,**” so your instructor can access them for grading without requesting individual access.



3. Prepare one detailed, professionally formatted PDF report covering Tasks 1 through 5, with a labelled screenshot and a short description for every step, plus the five video links.
4. From Task 2 onward, every terminal screenshot in the report must show your customized prompt from Task 3.
5. Submit the PDF report and the five video links through the Google Form the instructor will share separately, before the deadline.

10. Academic Integrity

This is individual work. Copying another student's terminal history, screenshots, explanations, or shell script, in whole or in part, is a violation of academic integrity, whether or not the wording is changed afterward. Submitted reports and scripts will be checked for similarity against other submissions and against public sources. Every explanation should be in your own words and reflect what you actually did on your own machine. Violations will be handled in accordance with AUCA's academic integrity policy.

11. Grading Rubric (100 Marks)

Assessment Component	Marks
Task 1: Environment setup, correctly installed and verified	10
Task 2: Student ID account created, used, and confirmed via whoami	15
Task 3: Terminal prompt correctly customized and used from this point onward	15
Task 4: Five videos, twelve commands each, handbook examples plus one original example per video	30
Task 5: Original, working, well-commented shell script using at least five commands	20
Report quality: clear screenshots, captions, explanations, and correct prompt throughout	10
Total	100

“Dream big and work hard to achieve your goals. Stay strong and keep fighting for your dreams. Do not let anything discourage you. The future looks bright.”

Good luck!